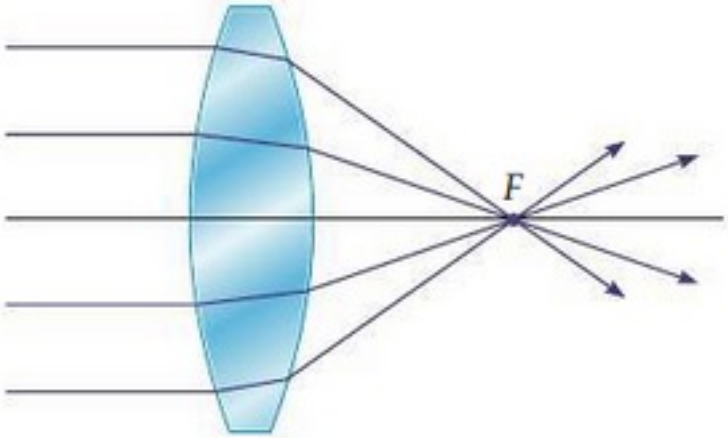
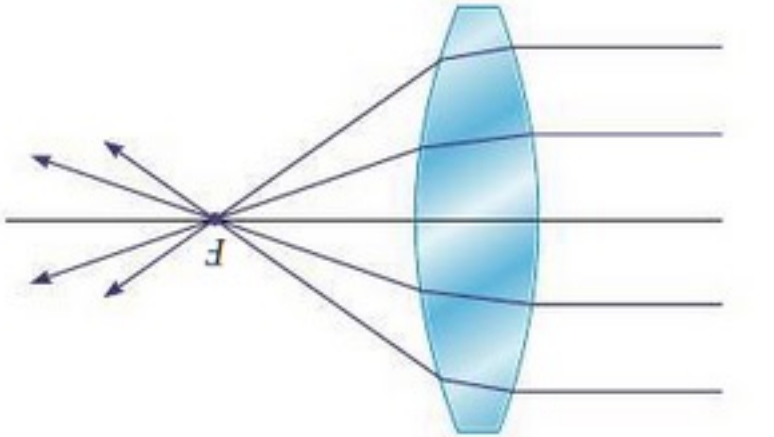


Lenses

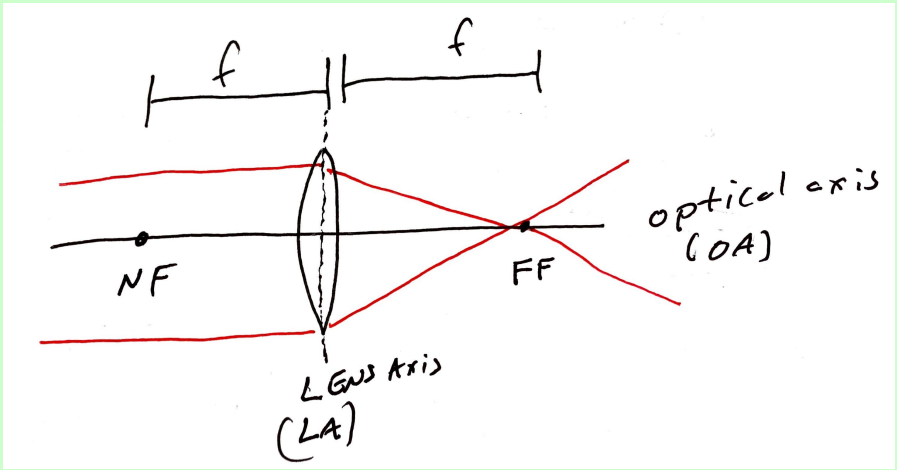
How a double convex lens works!!



(a) A double-convex lens



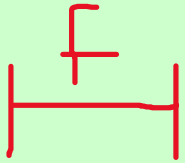
All lenses have two foci as light rays can be reversed

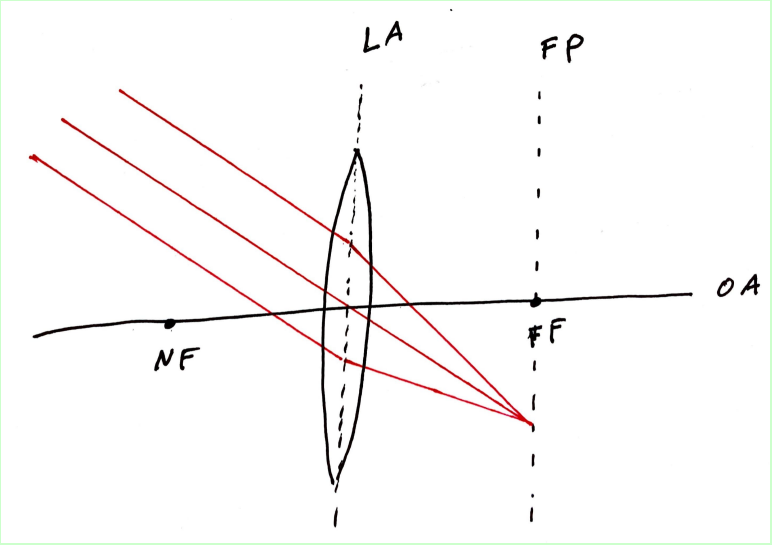


Usually called Near and Far Focus (NF and FF)

The two focal lengths are equal

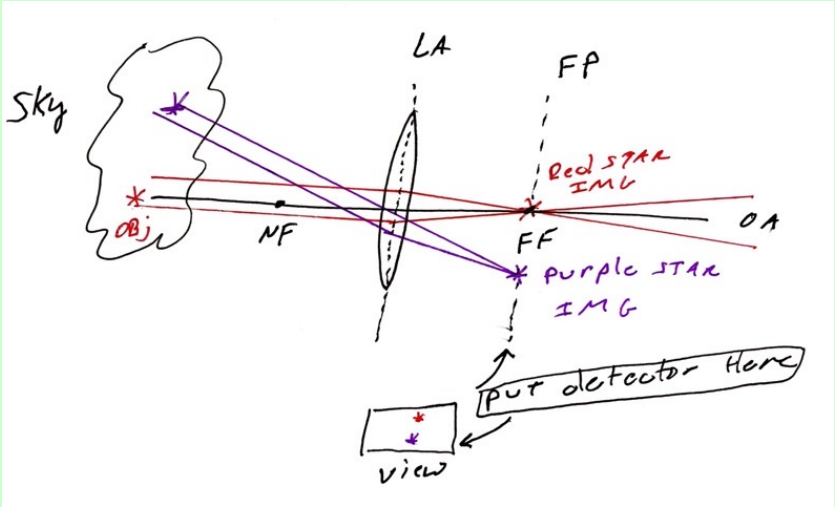
What if source is off axis?





Still focuses same distance away
But not on OA. Occurs on the
"Focal Plane"

What if lots of twinkly stars?

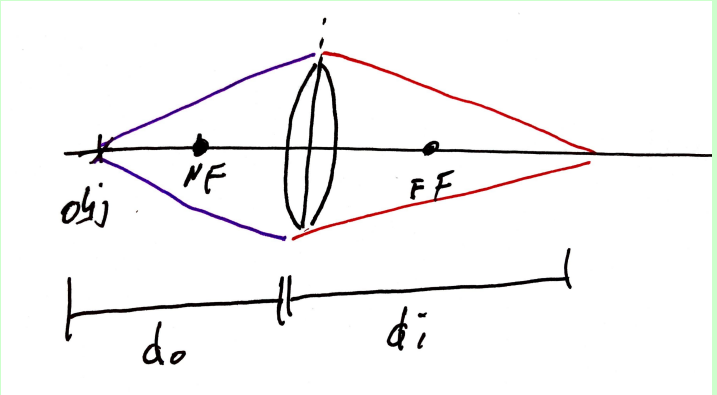


All the different sources focus at different parts of the focal plane. This is where your detector goes.

Thin Lens Eqn:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Familiar?



Focal length is something that changes do -> di

Shorter f means initial rays are bent more strongly.

Optometrists define the **power** of a lens, **P**

$$P = \frac{1}{f}$$

Ex. A 5D, or 5 Diopter lens means

$$P = 5 = \frac{1}{f}$$

$$f = 1/5 = 0.2 \text{ m} = \boxed{20 \text{ cm}}$$

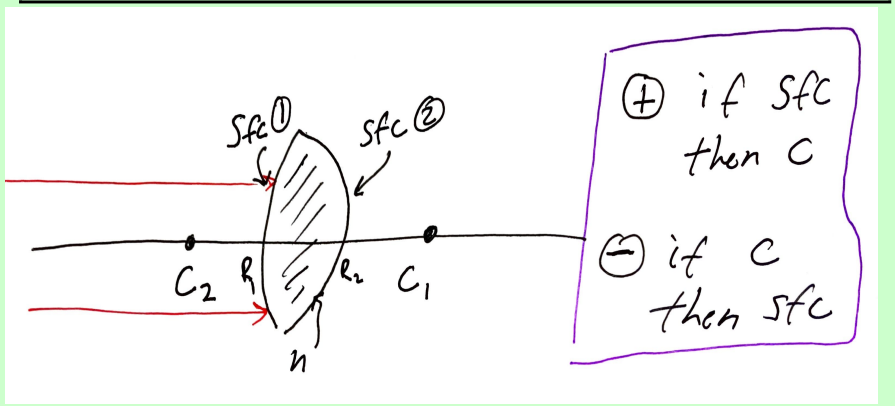
Lens Makers Equation

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For spherical mirrors: f = R/2, EASY

For Lenses, the equation is harder.

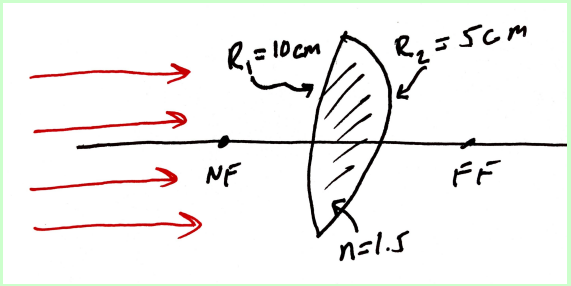
1. Two surfaces to deal with
2. Snell's law more complex than the law of reflection



Ex. Double convex lens. $|R_1|$ = 10 cm, $|R_2|$ = 5 cm, n = 1.5

$$\frac{1}{f} = (1.5-1)\left(\frac{1}{10} - \frac{1}{-5}\right)$$

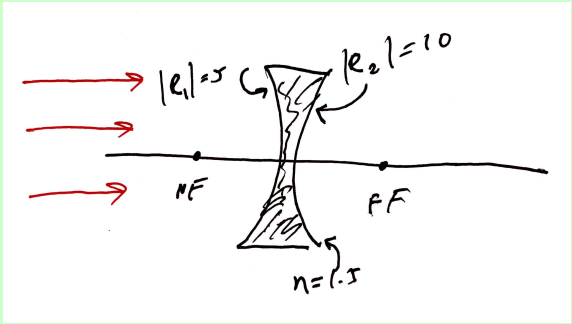
which means f=6.7 cm



Ex. Double concave lens. $|R_1|$ = 5 cm, $|R_2|$ = 20 cm, n = 1.5

$$\frac{1}{f} = (1.5-1)\left(\frac{1}{-5} - \frac{1}{10}\right)$$

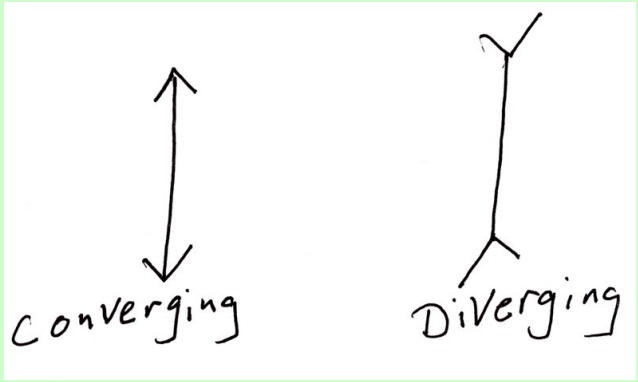
which means f=-6.7 cm



Converging lenses have + focal lengths Diverging lenses have - focal lengths

Ray Diagrams:

1. Draw OA, LA, NF, FF
2. Draw lenses as



3. Draw 3 Principal rays. They are slightly different depending on converging or diverging

Converging:

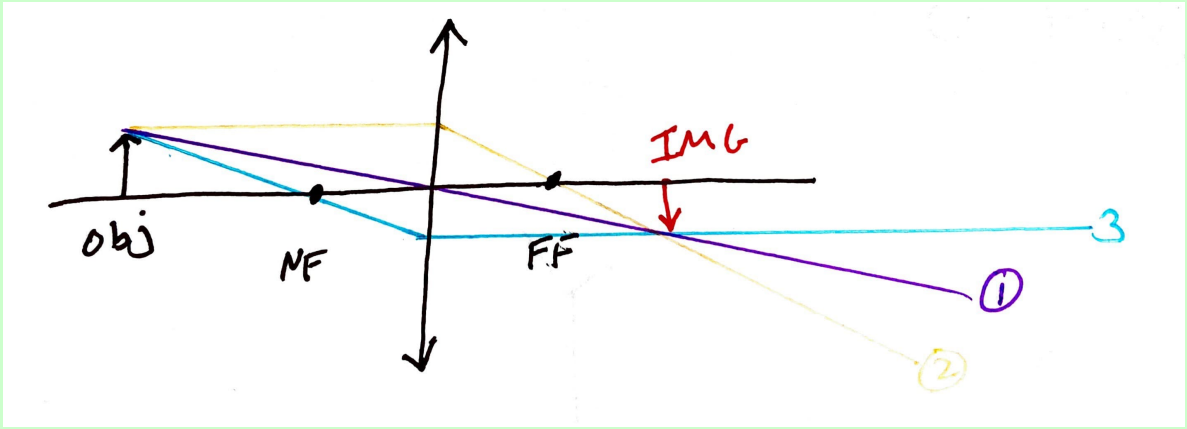
1. Thru the center of the lens and out ("run the gauntlet")
2. Parallel to OA -> Thru FF
3. Thru NF -> Parallel to OA

Diverging:

1. Thru the center of the lens and out ("run the gauntlet")
2. Parallel to OA -> Bend away from NF
3. Toward FF -> Parallel to OA

Ex. Converging lens with f = 5cm. do = 10 cm

A. Find IMG via ray tracing



B. Find via Thin-Lens Eqn: $\frac{1}{f} = \frac{1}{d_0} + \frac{1}{d_i}$

$$\frac{1}{5} = \frac{1}{10} + \frac{1}{d_i} \implies \boxed{d_i = 10 \text{ cm}}$$

Note: do + to left of lens
di + to right of lens

Lens sign conventions:

Quantity	Positive (+)	Negative (-)
d_o	left of lens	X
di	right of lens	left of lens
f	converging	diverging

Ex. Diverging diamond lens. n=2.4, $|R_1|$ = 15 cm, $|R_1|$ = 10 cm, do = 20

A. Use the lens makers equation to find f

B. Use Ray Tracing to find the IMG

C. Verify with the thin lens equation and definition of magnification